Globally Continuous Cohomology of Tori

A full answer to Question 10.1 of arXiv:1305.7269

Candidate solution packet for human review

13 August 2026

Abstract

Austin asked whether the globally continuous bar-cochain group $H^3_{\mathrm{cts}}(\mathbb{T}, \mathbb{T})$ is nonzero. We prove that it is zero. More generally, for the (necessarily trivial) continuous action of $\mathbb{T}^d$ on $\mathbb{T}^m$, we compute the whole theory:

$$H^n_{\mathrm{cts}}(\mathbb{T}^d, \mathbb{T}^m) \cong \begin{cases} \mathbb{T}^m, & n = 0, \\ \mathrm{Mat}_{m\times d}(\mathbb{Z}), & n = 1, \\ 0, & n \geq 2. \end{cases}$$

The proof avoids the unavailable long exact coefficient sequence. A normalized cochain in degree at least two factors through a simply connected smash product, hence lifts to a real cochain. For a cocycle the lift is a real cocycle, and Haar averaging supplies its primitive.

1 The published question

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, written additively. In Section 10.1 of [1], Austin considers the cohomology H^*_{cts} of the usual inhomogeneous bar complex, restricted to globally continuous cochains. The paper observes that this theory need not provide long exact sequences for general short exact sequences of topological modules, and asks:

Question 10.1. Is $H^3_{\mathrm{cts}}(\mathbb{T}, \mathbb{T})$ non-zero?

The question appears on physical page 91 of the source PDF:

10 Further questions

10.1 Continuous functions

All of our results have been about measurable functions $Z \longrightarrow A$. If one insists on continuous functions, then I do not know how to complete a similar analysis (unless A is a Euclidean space, for which easy arguments can then be made using Fourier analysis).

The problem is that our approach rests on reducing various calculations to cohomology, usually using long exact sequences. There is a cohomology theory for compact groups built using continuous cochains, $\mathrm{H}^*_{\mathrm{cts}}$, but it does not satisfy this axiom unless the ambient group Z is pro-finite. I do not know how to get around this difficulty; indeed, as an important consequence, I do not even know how to compute some quite simple instances of $\mathrm{H}^*_{\mathrm{cts}}$. Amazingly, the following seems to be open:

Question 10.1. *Is* $\mathrm{H}^3_{\mathrm{cts}}(\mathbb{T}, \mathbb{T})$ *non-zero?*

See [2] for more discussion of the defects of $\mathrm{H}^*_{\mathrm{cts}}$. Similar issues arise if one tries to work with smooth functions or other forms of regularity.

Austin–Moore define exactly this theory as continuous maps in the ordinary bar resolution [2, Section 2]. For a topological group G acting trivially on an abelian topological group A, put $C^n_{\text{cts}}(G, A) = C(G^n, A)$ and

$$\begin{aligned}(dc)(g_1, \ldots, g_{n+1}) = c(g_2, \ldots, g_{n+1}) \\ &+ \sum_{i=1}^{n} (-1)^i c(g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1}) \\ &+ (-1)^{n+1} c(g_1, \ldots, g_n).\end{aligned}$$

Then $H^n_{\text{cts}}(G, A) = \ker d / \operatorname{im} d$ in this complex.

2 Result and proof intuition

Theorem 1 (Full torus computation)**.** *Let $d, m \geq 1$. For the trivial action of $\mathbb{T}^d$ on $\mathbb{T}^m$,*

$$H^n_{\text{cts}}(\mathbb{T}^d, \mathbb{T}^m) \cong \begin{cases} \mathbb{T}^m, & n = 0, \\ \mathrm{Hom}_{\text{cts}}(\mathbb{T}^d, \mathbb{T}^m) \cong \mathrm{Mat}_{m \times d}(\mathbb{Z}), & n = 1, \\ 0, & n \geq 2. \end{cases}$$

In particular,

$$\boxed{H^3_{\text{cts}}(\mathbb{T}, \mathbb{T}) = 0,}$$

so the answer to Question 10.1 is no.

Idea of the proof. Normalize the bar complex. A normalized n-cochain vanishes whenever one input is the identity. For $n \geq 2$ it therefore factors through the smash product $(\mathbb{T}^d)^{\wedge n}$. This smash product is simply connected, so the cochain has a continuous logarithm $C : (\mathbb{T}^d)^n \to \mathbb{R}^m$. If the original cochain is a cocycle, dC is lattice-valued. Connectedness makes it constant, and normalization makes that constant zero. Thus C is an actual real cocycle. Finally, averaging its last variable against Haar measure gives an explicit real primitive. Reducing that primitive modulo $\mathbb{Z}^m$ completes the argument.

3 Three elementary lemmas

Lemma 2 (Continuous normalization). *The subcomplex of normalized continuous cochains*

$$N^n_{\mathrm{cts}}(G, A) := \{c \in C^n_{\mathrm{cts}}(G, A) : c(g_1, \dots, g_n) = 0 \text{ if some } g_i = e\}$$

has the same cohomology as $C^*_{\mathrm{cts}}(G, A)$.

Proof. This is the usual normalization theorem for the bar complex. Algebraically, the bar cochains form a cosimplicial abelian group; the normalized Moore complex is a deformation retract, while the complementary degenerate complex is contracted by the codegeneracies. In inhomogeneous coordinates those codegeneracies insert the identity in one coordinate. The projection and contracting homotopy are finite sums of coface and codegeneracy operators. All of these are compositions with continuous group operations or identity insertions, so they preserve continuity. Hence the standard algebraic normalization proof restricts verbatim to globally continuous cochains. □

Lemma 3 (Normalized torus cochains lift). *Let $n \geq 2$. Every normalized continuous map $c : (\mathbb{T}^d)^n \to \mathbb{T}^m$ has a normalized continuous lift $C : (\mathbb{T}^d)^n \to \mathbb{R}^m$ through the quotient map $q : \mathbb{R}^m \to \mathbb{T}^m$.*

Proof. Let $F_n \subseteq (\mathbb{T}^d)^n$ be the fat wedge, the union of the sets on which at least one coordinate is the identity. Since $c|_{F_n} = 0$, the map c factors as a based continuous map

$$\bar{c} : (\mathbb{T}^d)^n / F_n = (\mathbb{T}^d)^{\wedge n} \longrightarrow \mathbb{T}^m.$$

Give $\mathbb{T}^d$ its standard based finite CW structure. Every non-basepoint cell has dimension at least one, so every non-basepoint cell of the n-fold smash has dimension at least n. For $n \geq 2$ the smash product is connected and has no one-cells; hence it is simply connected. The universal covering $q : \mathbb{R}^m \to \mathbb{T}^m$ gives a unique based lift $\bar{C} : (\mathbb{T}^d)^{\wedge n} \to \mathbb{R}^m$. Pulling back gives C. Equivalently, the lift restricts on the connected fat wedge to a lattice-valued map taking the value 0 at the basepoint, so it is normalized. □

Lemma 4 (Haar contraction). *Let G be compact and let V be a finite-dimensional real vector space with trivial G-action. Every continuous V-valued cocycle in positive degree is a continuous coboundary.*

Proof. Let $C \in C^n_{\mathrm{cts}}(G, V)$ satisfy $dC = 0$, with $n \geq 1$, and let μ be normalized Haar measure. Define

$$B(g_1, \dots, g_{n-1}) = (-1)^n \int_G C(g_1, \dots, g_{n-1}, t)\, d\mu(t).$$

The function B is continuous. Integrate the identity $dC(g_1, \dots, g_n, t) = 0$ in t. Haar invariance changes the term containing $g_n t$ into the same integral defining $B(g_1, \dots, g_{n-1})$; collecting the remaining terms gives $dB = C$. □

4 Proof of the theorem

Proof of Theorem 1. In degree zero, the action is trivial, so the invariant group is $\mathbb{T}^m$. In degree one, the cocycle identity says precisely that $c(x+y) = c(x) + c(y)$. Thus the cocycles are continuous homomorphisms $\mathbb{T}^d \to \mathbb{T}^m$. Degree-one boundaries vanish for a trivial action, and the continuous homomorphisms between these tori are exactly the integer matrices $\mathrm{Mat}_{m\times d}(\mathbb{Z})$.

Now let $n \geq 2$ and let $c : (\mathbb{T}^d)^n \to \mathbb{T}^m$ be a continuous cocycle. By Lemma 2, we may assume c is normalized. By Lemma 3, choose a normalized continuous lift $C : (\mathbb{T}^d)^n \to \mathbb{R}^m$ with $q \circ C = c$.

Since $dc = 0$, the map

$$dC : (\mathbb{T}^d)^{n+1} \longrightarrow \mathbb{R}^m$$

takes values in the deck lattice $\mathbb{Z}^m$. Its domain is connected, so dC is constant. At the all-identity tuple every term in dC is a value of the normalized cochain C on a tuple containing an identity; all those terms are zero. Hence $dC = 0$ everywhere.

Lemma 4 now gives a continuous cochain $B : (\mathbb{T}^d)^{n-1} \to \mathbb{R}^m$ with $dB = C$. Therefore

$$c = q \circ C = d(q \circ B),$$

so c is a continuous coboundary. This proves the asserted vanishing for all $n \geq 2$. □

Remark 5 (Module action)**.** *There is no omitted nontrivial toral action in Theorem 1. The group of continuous automorphisms of $\mathbb{T}^m$ is $\mathrm{GL}(m,\mathbb{Z})$ with the discrete topology. Any continuous action by automorphisms corresponds to a continuous homomorphism $\mathbb{T}^d \to \mathrm{GL}(m,\mathbb{Z})$, which is trivial because $\mathbb{T}^d$ is connected.*

Remark 6 (Why this does not misuse a long exact sequence)**.** *The source emphasizes that the presentation $\mathbb{Z}^m \to \mathbb{R}^m \to \mathbb{T}^m$ need not yield a long exact sequence for globally continuous cochains, because a general continuous torus-valued cochain need not admit a global continuous lift. The proof above makes no such assertion. It first normalizes a cochain of degree at least two; only then does the fat-wedge factorization prove that this particular cochain has a lift. The cocycle equation is then checked again upstairs. Thus the precise defect highlighted in [1, 2] is bypassed, not assumed away.*

5 Independent winding-number audit

There is a direct degree-three check. A map $\mathbb{T}^n \to \mathbb{T}$ has a winding vector in $\mathbb{Z}^n$. The bar differential induces

$$D_2(b_1, b_2) = (-b_1, 0, b_2), \qquad D_3(a_1, a_2, a_3) = (0, a_2, a_2, 0).$$

Consequently

$$\ker D_3 = \{(a, 0, c) : a, c \in \mathbb{Z}\} = \operatorname{im} D_2.$$

Thus every degree-three cocycle has coboundary winding, after which it lifts to $\mathbb{R}$ and the proof above applies. The accompanying exact-arithmetic script checks these matrices, the bar signs, and $D_{n+1}D_n = 0$ through degree eight. This calculation is only a sanity check; the normalized smash-product proof already handles all degrees and all finite-dimensional tori.

6 Scope, novelty, and review status

Theorem 1 fully answers the exact Question 10.1 and strengthens it to all degrees and all finite-dimensional source and coefficient tori. It does not complete Austin's broader program for continuous partial-difference equations. Nor does it assert vanishing for measurable, locally continuous, smooth-local, Segal–Mitchison, or classifying-space cohomology; those are different theories and may be nonzero [2, 3].

A bounded literature search on 13 August 2026 used the exact question and notation, close variants with "globally continuous cohomology", circle and torus coefficients, the source title/id/author, and citation searches. The primary sources [2, 3] discuss the same globally continuous theory but do not state this computation. No later explicit answer or equivalent full characterization was found. Novelty confidence is moderate: the argument may be implicit folklore under different notation, but the exact published question and bounded search revealed no answer.

The result is marked *candidate full solution, likely valid*. Human review should focus on Lemmas 2 and 3; once those standard facts are accepted, the remaining proof is an explicit cocycle and Haar-integration calculation.

References

[1] T. Austin, *Partial difference equations over compact Abelian groups, I: modules of solutions*, arXiv:1305.7269, 2013; Question 10.1 on physical PDF page 91.

[2] T. Austin and C. C. Moore, *Continuity properties of measurable group cohomology*, arXiv:1004.4937, 2010.

[3] F. Wagemann and C. Wockel, *A cocycle model for topological and Lie group cohomology*, arXiv:1110.3304, 2011.